

SETS

CSE 2213 – Discrete Mathematics

WE KNOW ABOUT SETS,
RIGHT?

SETS

A set is a collection of unordered objects.

- $V = \{a, e, i, o, u\}$ is the set of all English vowels
 - $a \in V$ means a is a member of V
- $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ is the set of all positive integers less than or equal to ten

SOME IMPORTANT SETS

- $N = \{0, 1, 2, 3, \dots\}$ is the set of all natural numbers
- $Z = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is the set of all integers
 - $Z^+ = \{1, 2, 3, \dots\}$ is the set of all positive integers
- $Q = \left\{ \frac{p}{q} \mid p \in Z, q \in Z \text{ and } q \neq 0 \right\}$ is the set of all rational numbers
 - Q^+ is the set of all positive rational numbers
- R is the set of all real numbers

SET BUILDER NOTATION

- $S = \{1,2,3,4,5,6,7,8,9,10\}$ is the set of all positive integers less than or equal to ten
 - **Set builder notation:** $S = \{x \in \mathbb{Z}^+ \mid x \leq 10\}$
 $S_1 = \{x \in \mathbb{Z}^+ \mid x \text{ is even and } x \leq 10\}$
 $S_2 = \{x \in \mathbb{Z}^+ \mid x \text{ is prime and } x \text{ is divisible by } 3\}$

SUBSETS

- *If every element of one set is present in another set, then the first set is a subset of the second set. Example:*
- *Let $U = \{1, 2, 3, 4\}$*
- *$V = \{2, 3, 4, 1, 6\}$*
- We can see that every element of U is present in V . Therefore, U is a subset of V
- $U \subseteq V$ [This is the subset symbol]
- If U is a subset of V , AND U and V are not exactly the same sets, then U is a proper subset of V . This can happen if V contains at least one element that is not present in U .
- In our example, U is a proper subset of V
 - $U \subset V$ [This is the proper subset symbol]

EMPTY SET

- $\emptyset$
- $\{\}$
- $\{\emptyset\}$

CARDINALITY OF FINITE SETS

- Finite set: A set S with exactly n number of distinct elements, where $n \in \mathbb{Z}$ and $n \geq 0$
 - $|S|$ denotes the cardinality or size of set S
 - $n = |S|$
- $V = \{a, e, i, o, u\}$
 - Cardinality of V , $|V| = 5$
- The cardinality of the empty set is 0.
 - That is, $|\emptyset| = 0$

POWER SET

- Power set of S , $P(S) = \{A \mid A \subseteq S\}$
 - It is a set of sets
 - The power set of S consists of all possible subsets of S
- $\emptyset \subseteq S$
- $S \subseteq S$
- Example: What is the Power Set of $\{1, 2\}$?
 - $P(\{1, 2\}) = \{\{1\}, \{2\}, \{1, 2\}, \emptyset\}$
 - What is the cardinality of $P(\{1, 2\})$?
 - In general, what is the cardinality of $P(S)$?
 - $2^{|S|}$

CARTESIAN PRODUCT

- The Cartesian product of two sets A and B , denoted $A \times B$, is the set of all possible **ordered tuples of the form (x, y)** where x is an element of A and y is an element of B .
- $A \times B = \{(x, y) \mid x \in A \wedge y \in B\}$
- For example:
- $A = \{1, 2, 3\}, B = \{a, b\}$
 $\therefore A \times B = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$
- Note that, in $A \times B$, elements of A appears first, and elements of B appear next. This ordering is important
- What would be the Cartesian Product of $B \times A$?

CARTESIAN PRODUCT

- $A = \{1,2\}, B = \{a,b\}, C = \{x,y\}$
Find out $A \times B \times C$
- $\{(1, a, x), (1, a, y), (1, b, x), (1, b, y), (2, a, x), (2, a, y), (2, b, x), (2, b, y)\}$
- General formula for the cardinality of Cartesian Product Set?
- $|A \times B| = |A| * |B|$